

FARIS MUNIR MAHDI

Software Engineer | Backend & Systems Architecture

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PROFESSIONAL SUMMARY

Software Engineer focused on backend and systems engineering, with production experience across real-time applications, media-processing pipelines, IoT services, and containerized deployments. Builds Rust and TypeScript services backed by PostgreSQL, connects web and mobile clients to asynchronous workflows, and integrates practical AI capabilities with attention to authorization, observability, and operational boundaries.

SKILLS

Programming	Rust, Go, TypeScript, JavaScript, Python, C++, PHP, Kotlin, SQL, Bash, QML
Web & Mobile	Svelte, Vue, Nuxt, React, HTML, CSS, Android
Backend	Axum, NestJS, FastAPI, Hapi, REST APIs, WebSockets, MQTT
Data & Storage	PostgreSQL, MySQL, Redis, TimescaleDB, pgvector
Infrastructure	Docker, Linux, GitHub Actions, OpenTelemetry, Prometheus, Grafana
AI Systems	LangGraph, RAG, MCP, Ollama

PROFESSIONAL EXPERIENCE

Backend & Fullstack Developer | PT Perkasa Pilar Utama | July 2025 — Present | Jakarta, Indonesia

Develops backend services, client integration, and production operations for meeting-processing and IoT products, covering asynchronous workflows, real-time state, authorization, and on-premise delivery.

Related projects: Sensio Notes: Meeting Intelligence Platform, Sensio IoT: Smart-Space Platform

- Built the Sensio Notes backend flow for long meeting recordings with PostgreSQL data models, chunked S3 uploads, and background transcription; exposed WebSocket progress updates so clients can follow processing while work continues asynchronously.
- Connected transcripts to LangGraph workflows that produce reviewable meeting notes and action items, integrating AI-assisted processing into the application flow and keeping the outputs available for user review.
- Built and maintained Sensio IoT services in Rust and NestJS, enforcing site- and room-scoped access before device commands and connecting telemetry ingestion with MQTT and Zigbee2MQTT for authorized device interaction.
- Integrated web and mobile clients with backend state, traced failures with OpenTelemetry and structured logs, and packaged services with Docker for on-premise operation; followed issues from development through production delivery.

SEO Specialist | PT Tradeasia International Indonesia | Jan 2024 — Mar 2024

Worked on search visibility through keyword research, on-page SEO, technical checks, and traffic analysis.

- Researched search queries, reviewed page structure and technical SEO issues, and used analytics data to decide which pages and topics needed attention.

SELECTED PROJECTS

Masih Awam AI Code: Agentic Coding Workspace | 2026

Built a coding workspace that connects repository inspection, file changes, command execution, and verification through bounded capabilities. Separated agent reasoning from execution authority with Rust, MCP, and Bubblewrap so repository operations pass through explicit policy boundaries, while the workspace exposes progress and evidence from the actual development environment.

Masih Awam LMS: Game-Based Learning Platform | 2026

Built Rust and Axum services backed by PostgreSQL for courses, enrollment, and lesson completion. Separated auditable progress state from worlds, quests, and story presentation, allowing learner dashboards to show the next step while public course pages and private progress views follow different data ownership and rendering boundaries.

IoTNet: IoT Operations Platform | 2026

Built an IoT management platform using TypeScript, Hapi, PostgreSQL, MQTT, and EMQX to connect user workflows with asynchronous device communication. Separated requested commands from observed device state and checked ownership before control, giving the application a clear model for device operations, automation decisions, and operational visibility through OpenTelemetry.

AskIn: AI Workspace | Ruang Guru Academy | 2024

Adapted the backend and integration layer of an Open WebUI-based application during Ruangguru Academy. Worked with FastAPI request handling, authenticated sessions, model discovery, Ollama-compatible and OpenAI-compatible routing, and retrieval context to connect the Svelte chat interface with configurable model services and Docker-based deployment.

TECHNICAL PROGRAMS

Machine Learning Engineer | DBS Foundation | Feb 2025 — July 2025

Related projects: GiziLens: A Team Capstone Around Food Recognition

- Contributed to the application side of the GiziLens team capstone, implementing authentication, sessions, and user APIs around a food-recognition subsystem developed within the team.
- Integrated PostgreSQL and Redis with the Nuxt frontend and containerized services, providing a deployable web application around the team's food-recognition subsystem.

Full Stack Web Developer | Codepolitan | Sep 2024 — Dec 2024

- Worked through backend development with Node.js and Express.js, including API design and application data stored in MongoDB.

Back End Developer | Ruang Guru Academy | Feb 2024 — Jun 2024

Related projects: AskIn: AI Workspace

- Worked through backend-focused learning and implementation around application APIs, service integration, data flow, and deployment.

Cloud Computing Cohort | Bangkit Academy | Aug 2023 — Jan 2024 | Remote · Indonesia

Related projects: Parentify: Parenting Companion

- Completed the Cloud Computing learning path through self-paced coursework, instructor-led sessions, hands-on cloud labs, English practice, and soft-skill training.
- Worked in the Parentify cross-functional capstone team where technical decisions had to be communicated clearly across Cloud Computing, Mobile Development, and Machine Learning roles.

EDUCATION

UPN "Veteran" East Java | Bachelor of Informatics | 2021 — 2025

CERTIFICATIONS

Alibaba Cloud Certified Professional | Alibaba Cloud

Google IT Support Professional Certificate | Coursera | Sep 2023

Architecting on AWS (Membangun Arsitektur Cloud di AWS) | Dicoding